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Massachusetts' 2026 Clean Peak Standard Recalibration

Adaptation and storage tradeoffs

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Joint work with Boyu Yao, Johns Hopkins University

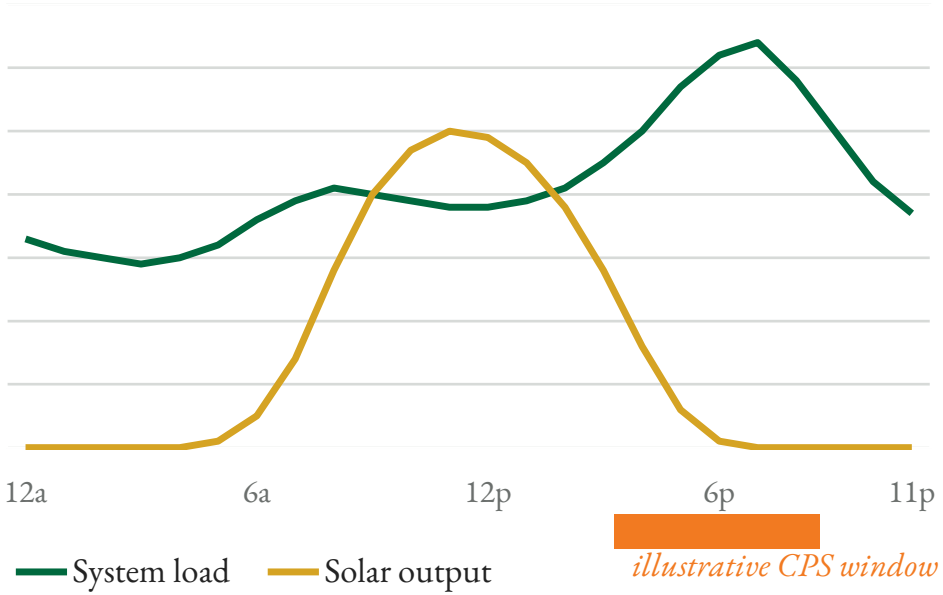
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Clean Peak Standard (CPS) adds time to Massachusetts clean energy policy

Why timing matters



Solar output can decline before the evening system peak, so annual clean energy totals do not ensure clean supply in high-value hours.

What the CPS does

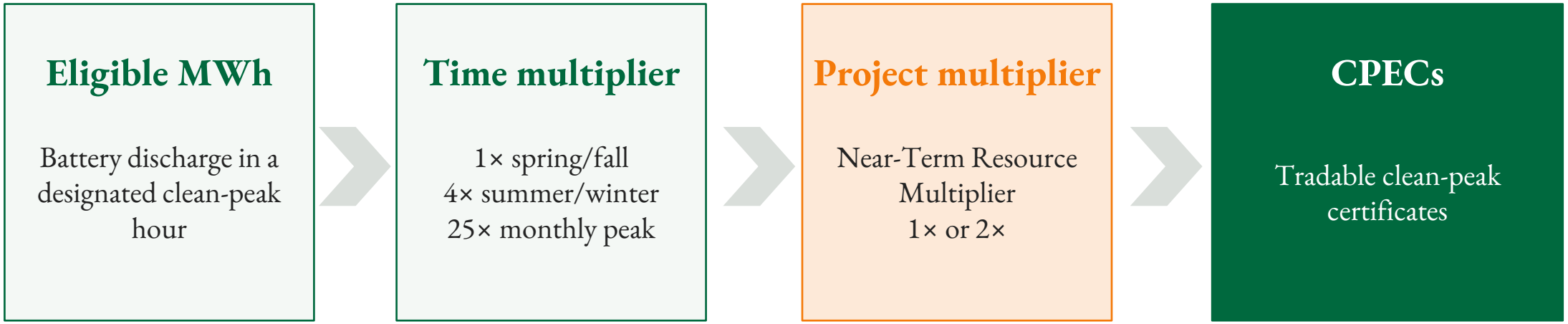
Massachusetts requires retail electricity suppliers to acquire Clean Peak Energy Certificates (CPECs) equal to a percentage of annual sales.

- **Earn CPECs in designated peak periods**
  Eligible generation, battery discharge, and demand reduction receive certificates only during clean-peak hours.
- **Weight higher-value delivery**
  Seasonal, hourly, monthly-peak, and project multipliers increase certificates per physical MWh.
- **Meet an annual supplier obligation**
  Required CPECs = Minimum Standard × retail sales; shortfalls can be covered through the $65/CPEC ACP.

This paper asks: When the 2026 rule lowers CPEC demand and expands a 2× storage multiplier, what happens to investment, discharge, cost, and fossil generation?

Source: Massachusetts DOER Clean Peak Energy Standard; Dai & Yao (2026).

CPS rewards discharge in high-value hours



Supplier obligation Minimum Standard × annual retail electricity sales

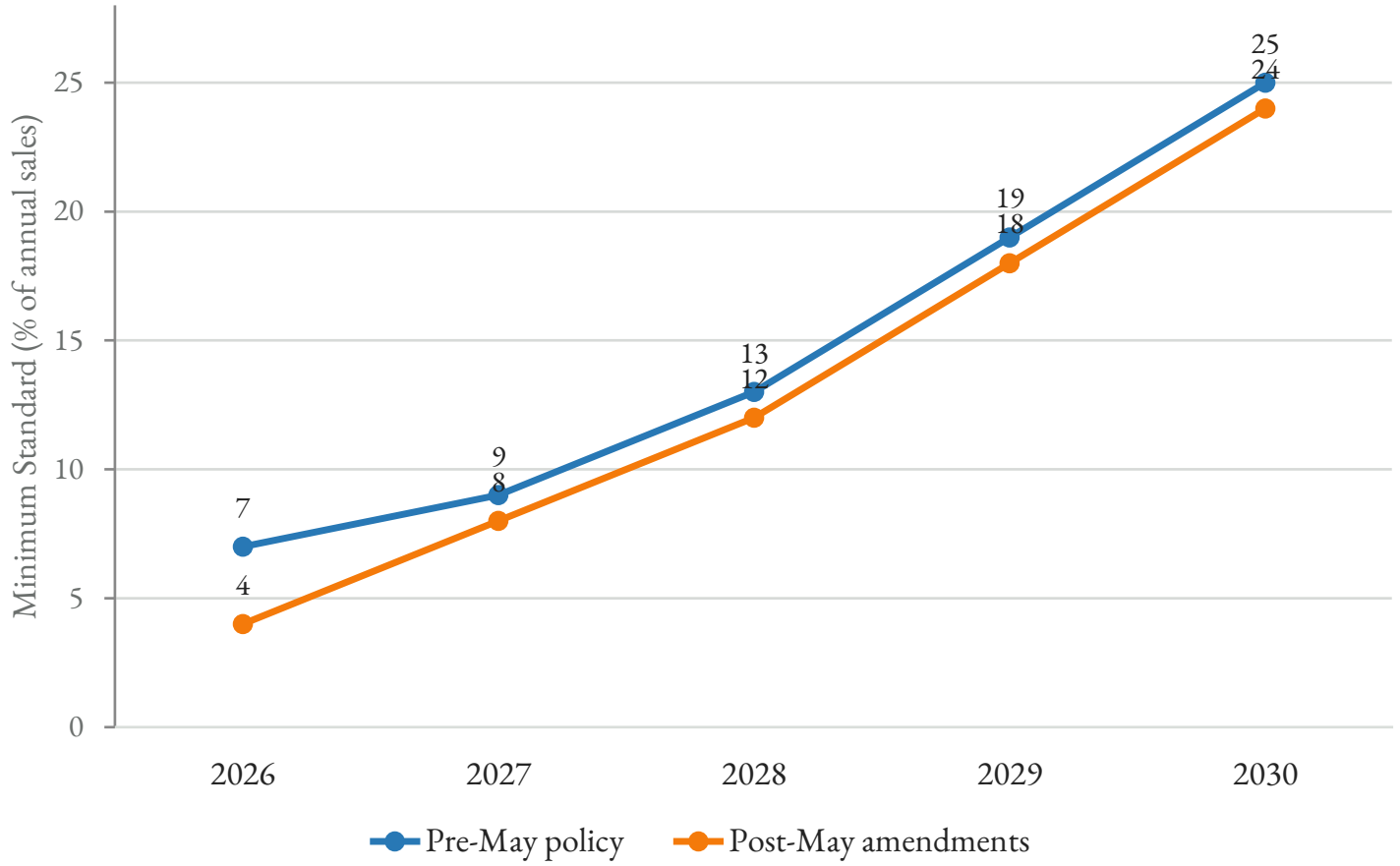
Any shortfall can be covered by the Alternative Compliance Payment: $65 per CPEC.

Key distinction

A multiplier increases certificates per MWh; it does not directly create more battery capacity or physical discharge.

Source: Massachusetts DOER Clean Peak Energy Standard; 225 CMR 21.

2026 recalibration: lower standards, 2× NTRM



Near-Term Resource Multiplier

1× → 2×

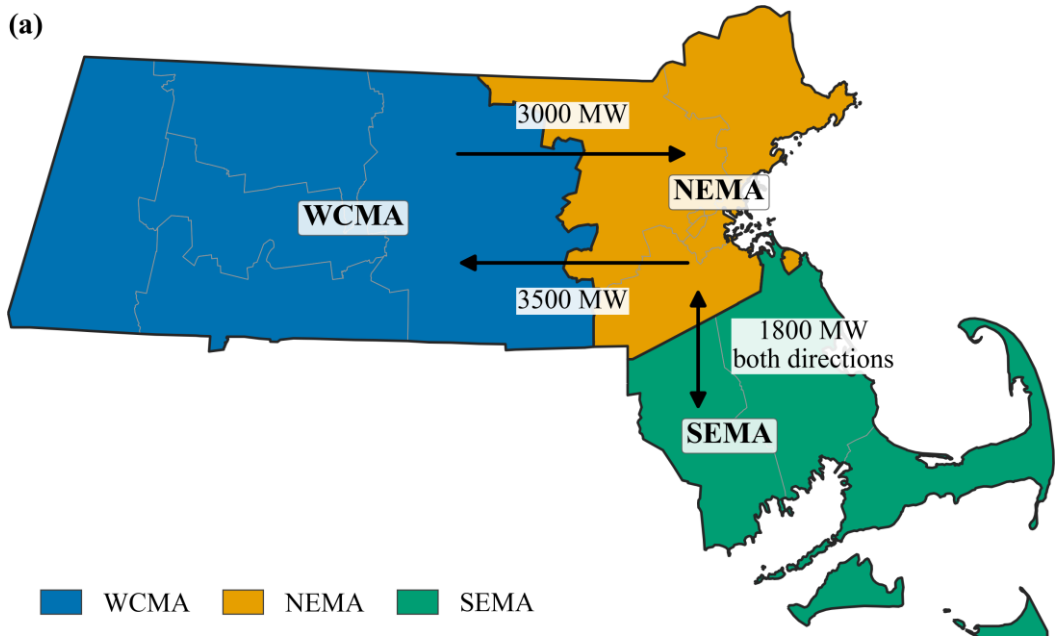
qualifying candidate storage

The recalibration changes both sides of the certificate market:

- less aggregate CPEC demand
- more CPECs per eligible MWh

Source: Massachusetts DOER emergency rulemaking filed May 28, 2026; Dai & Yao (2026).

Three-zone hourly model and policy scenarios



System and horizon

- Three Massachusetts zones with directional transfer limits
- 2026–2030; 8,760 chronological hours in each year
- Endogenous four-hour batteries and offshore wind
- Hourly balance, resource adequacy, CPS, and RGGI cap

Policy cases

Pre-May higher standards · NTRM = 1×

Post-May revised standards · NTRM = 2×

Both scenarios achieve full CPS compliance, with zero ACP, load shedding, or resource-adequacy shortfall.

Source: Dai & Yao (2026), model formulation and data assumptions.

Optimization model and policy constraints

Objective

$$\min \sum_{y \in \mathcal{Y}} \left(C_y^{\text{in}} + C_y^{\text{op}} + C_y^{\text{pe}} \right)$$

Annualized investment + fixed and variable operations + policy and reliability penalties

Hourly physical system

Zonal balance

$$\sum_{g \in \mathcal{G}_i} p_{gthy} + \sum_{g \in \mathcal{G}_i^{\text{st}}} (dc_{gthy} - c_{gthy}) - \sum_{l \in \mathcal{LS}_i} q_{lthy} + \sum_{l \in \mathcal{LR}_i} q_{lthy} = L_{ithy} - p_{ithy}^{\text{LS}}$$

Generation, storage, and transfers meet hourly net load in each zone.

Storage chronology

$$e_{gthy}^{\text{SOC}} = e_{g,\pi(t,h),y}^{\text{SOC}} + \eta_g^{\text{CH}} c_{gthy} - \frac{dc_{gthy}}{\eta_g^{\text{DC}}}$$

State of charge is linked across all 8,760 hours in each model year.

Policy constraints

Clean Peak Standard

$$\sum_{g,t,h} N_{ty} M_{thy}^{\text{CPS}} M_{gy}^{\text{NTRM}} dc_{gthy} + e_y^{\text{CPS}} \geq R_y^{\text{CPS}} \sum_{i,t,h} N_{ty} L_{ithy}$$

Multiplier-adjusted battery discharge must meet the annual CPEC requirement.

NTRM scales certificates, not storage or discharge.

Resource adequacy

$$\sum_k \sum_{g \in \mathcal{G}^k} \overline{P}_g^{\text{G}} F_{ky}^{\text{ELCC}} o_{gy} + p_y^{\text{RA}} \geq (1 + R_y^{\text{RA}}) \overline{L}_y$$

ELCC-adjusted capacity must cover peak load plus the reserve margin.

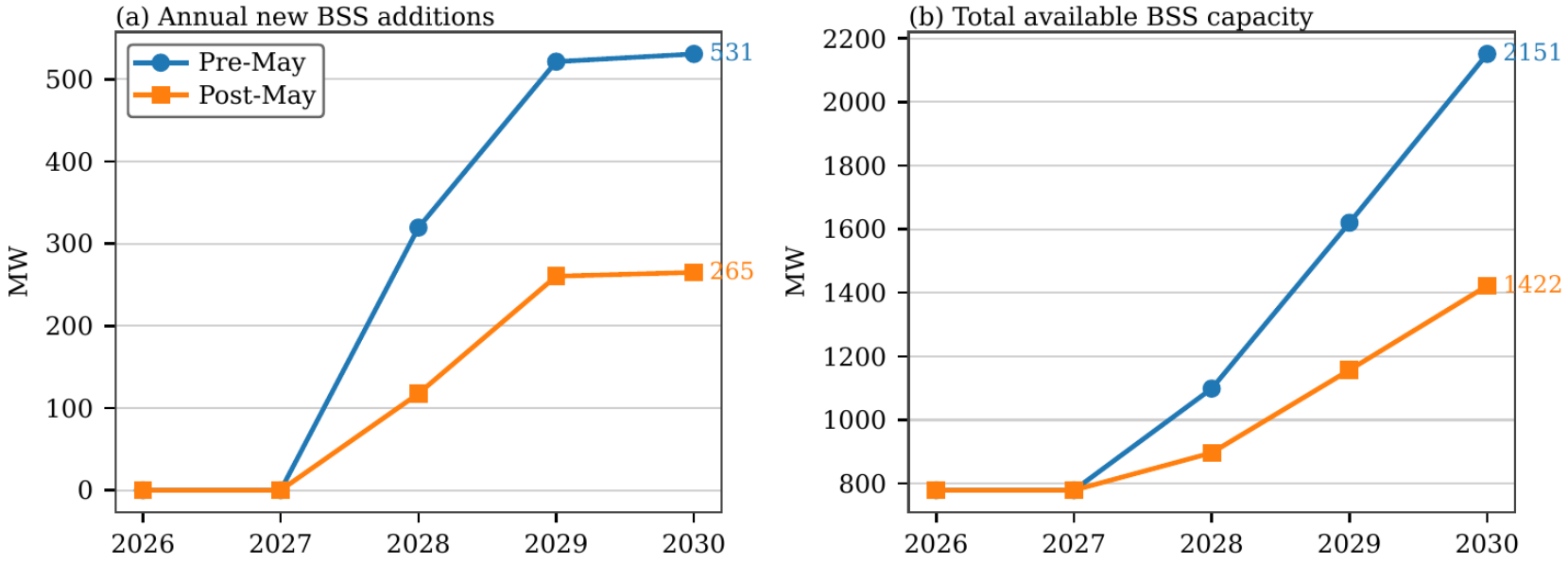
RGGI emissions cap

$$\sum_{g,t,h} N_{ty} e_g^{\text{CO}_2} p_{gthy} \leq \overline{E}_y^{\text{CO}_2}$$

Model simplifications: Continuous capacity expansion · Four-hour batteries · Fixed existing fleet · No unit commitment

Source: Dai & Yao (2026), objective and policy constraints in the capacity-expansion model.

Post-May battery additions fall 53%



Cumulative new battery capacity

1.372 GW

↓

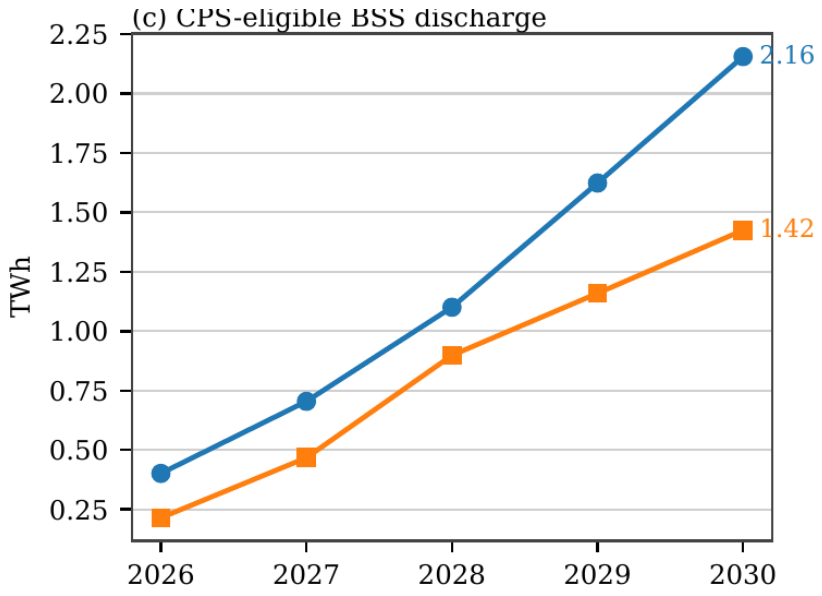
0.643 GW

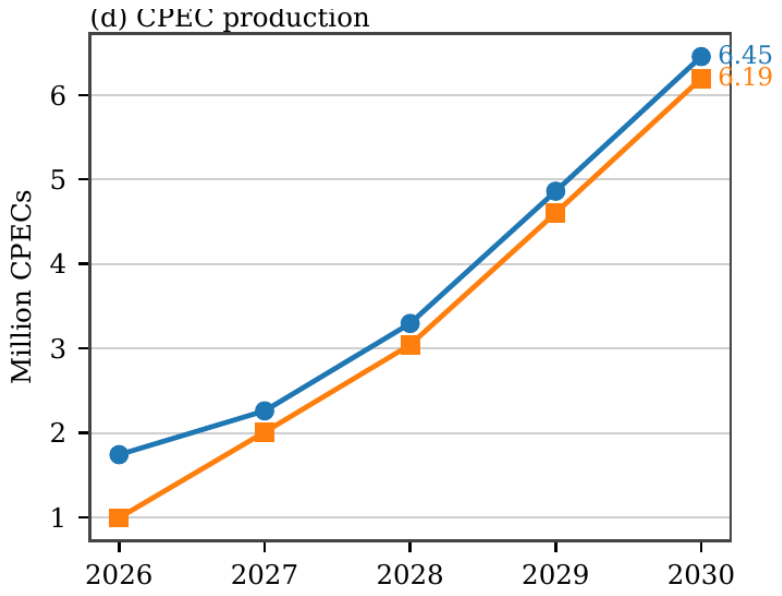
−53.1%

2030 total capacity
2.151 → 1.422 GW

Source: Dai & Yao (2026), model results for 2026–2030.

CPEC output falls less than physical discharge





Physical delivery

5.986 → 4.165 TWh

−30.4%

Multiplier-adjusted accounting

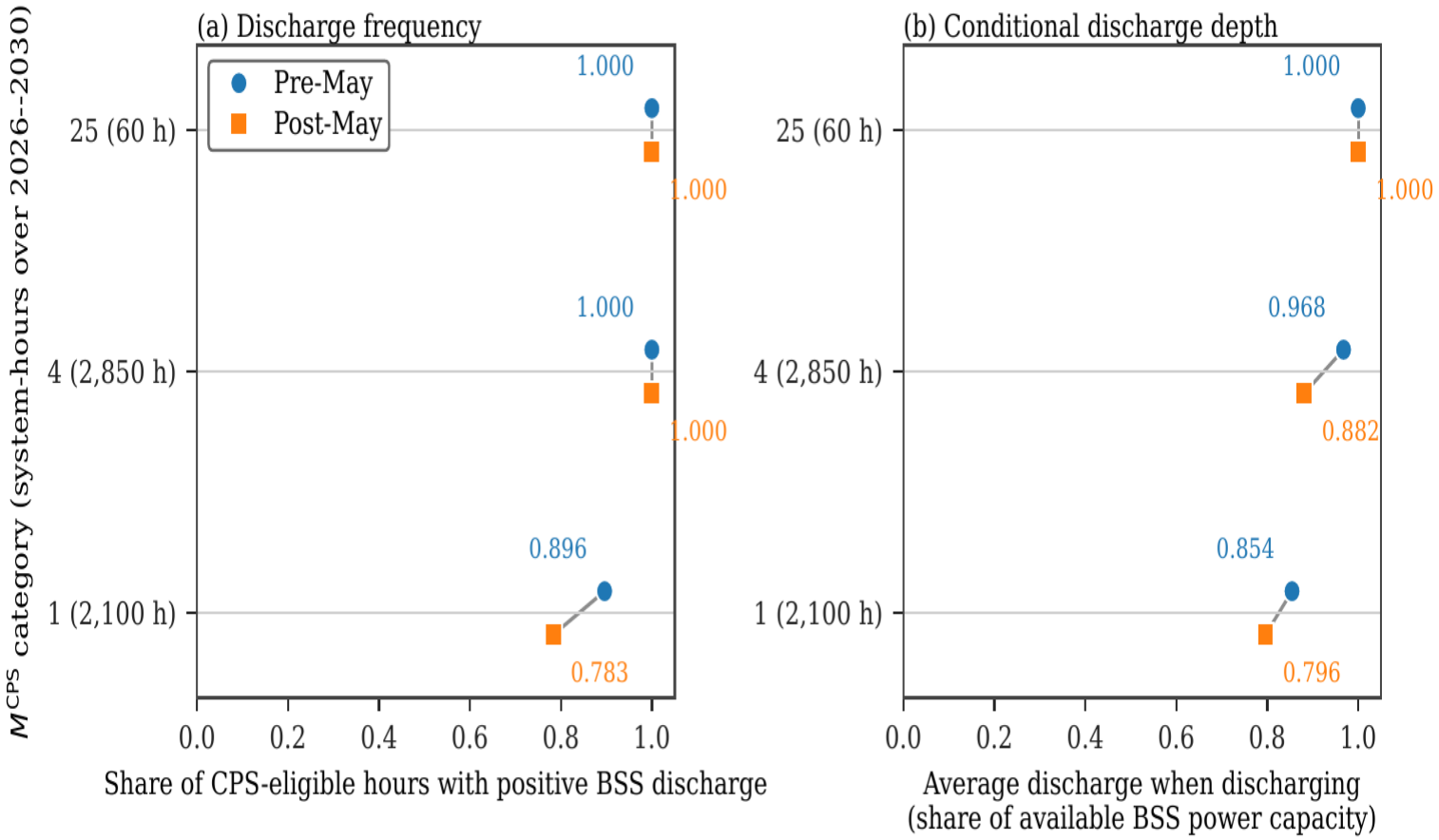
18.613 → 16.848 million CPECs

−9.5%

The 2× NTRM preserves certificates more effectively than it preserves physical discharge.

Source: Dai & Yao (2026), cumulative model outcomes for 2026–2030.

Peak-hour response survives; broader discharge weakens



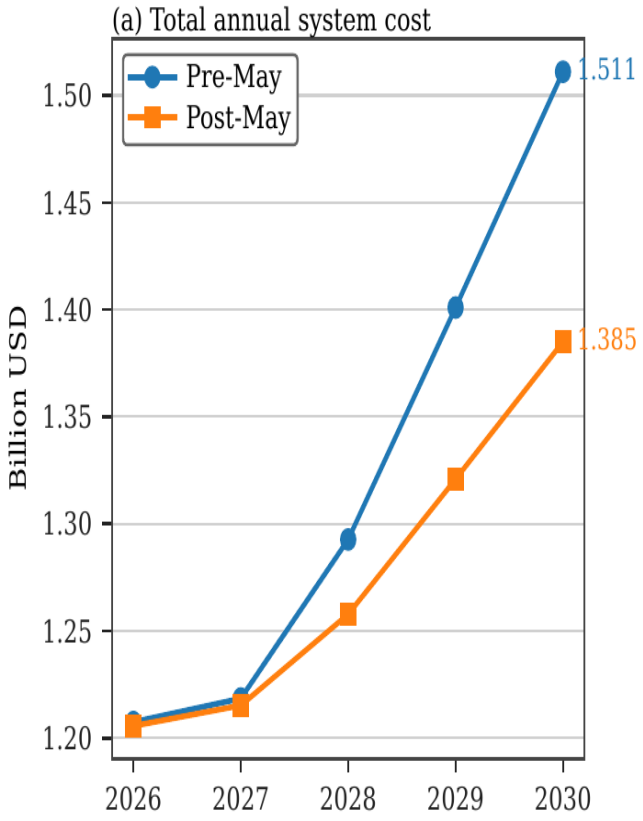
How operation changes

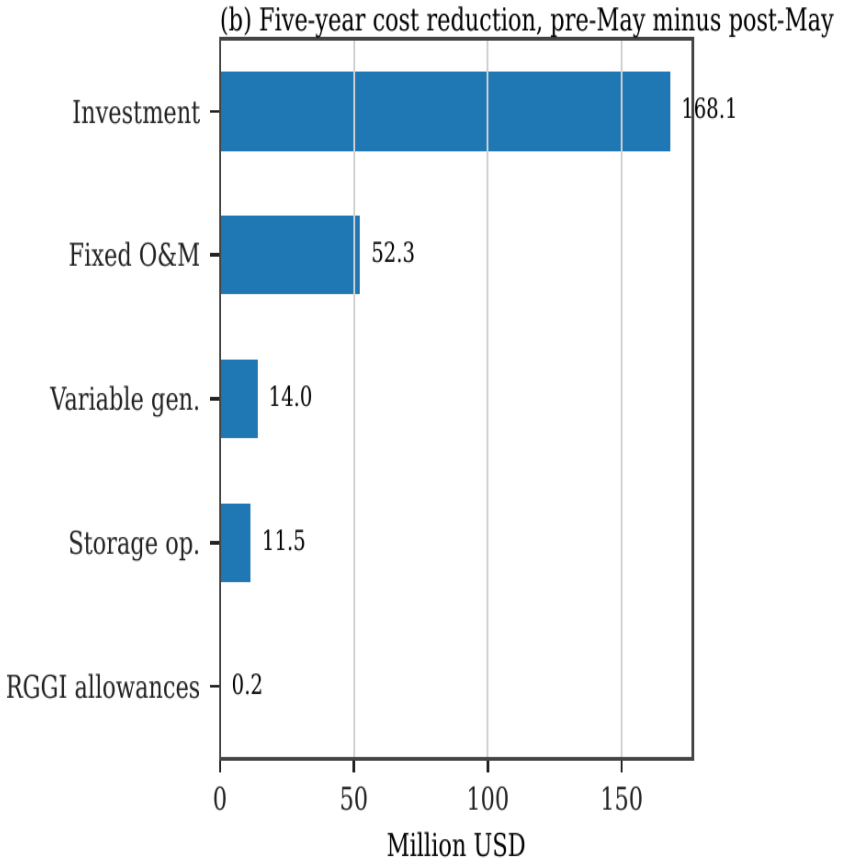
- **25× monthly peak**
  Frequency and discharge depth remain 1.000 in both policy cases.
- **4× seasonal hours**
  Frequency remains 1.000, but conditional depth falls from 0.968 to 0.882.
- **1× hours**
  Frequency falls 0.896 → 0.783; conditional depth falls 0.854 → 0.796.

The strongest peak signal is preserved, while response in the many lower-multiplier hours becomes less frequent and less deep.

Source: Dai & Yao (2026). Frequency and depth are computed across CPS-eligible system-hours, 2026–2030.

Avoided battery buildout explains most cost savings





Five-year system cost reduction

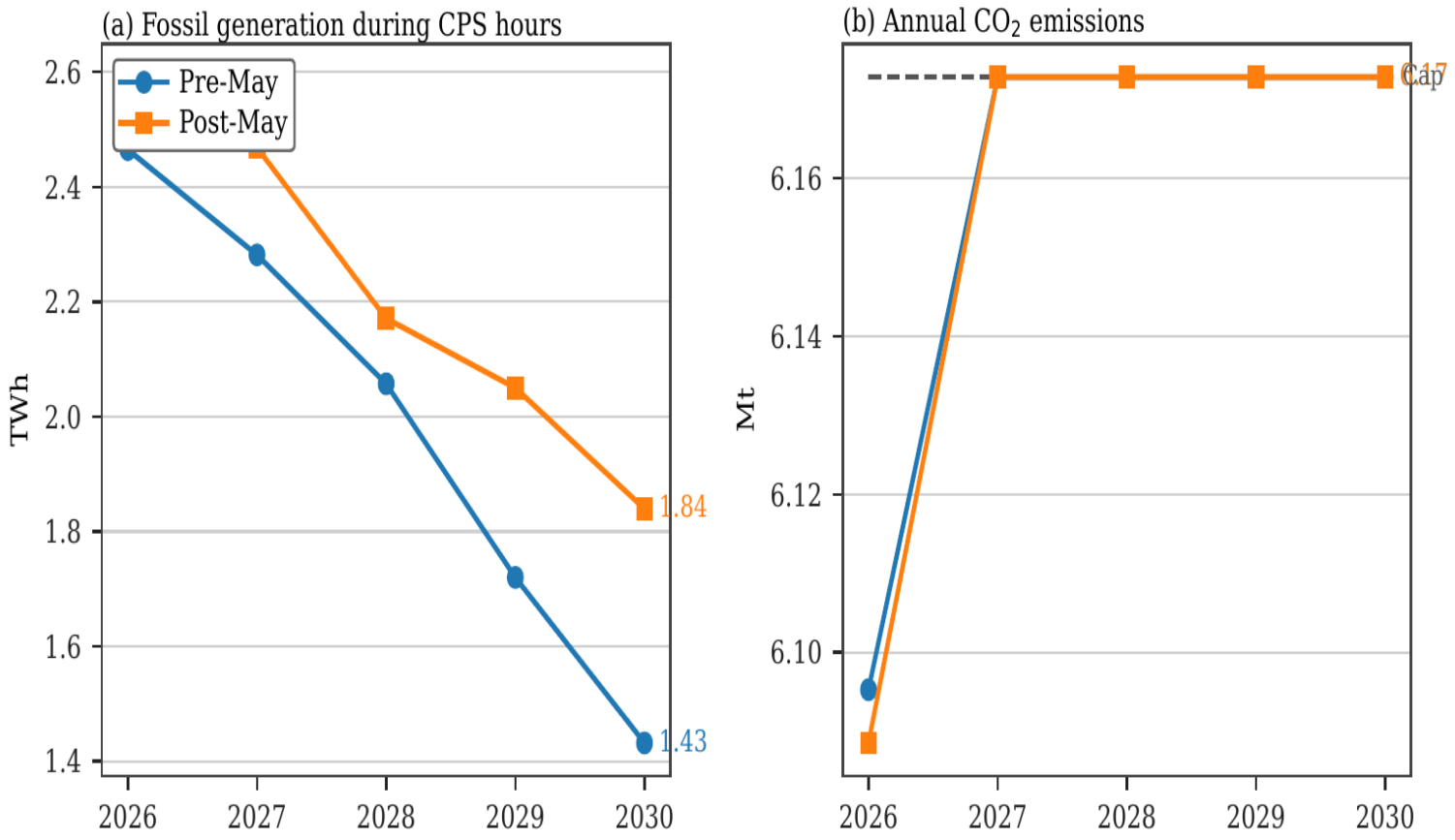
$246.1M **3.7%**

- **Investment: $168.1M**
  About 68% of total savings comes from avoided battery capital.
- **Fixed O&M: $52.3M**
  Another 21% comes from lower recurring fixed costs.
- **Other components: $25.7M**
  Variable generation, storage operation, and RGGI account for the remaining 10%.

The cost gap widens after 2028, when battery buildout begins to diverge.

Source: Dai & Yao (2026), five-year modeled system costs, 2026–2030.

More fossil generation shifts into CPS hours



Policy interaction

- **CPS-hour fossil output rises**
  Cumulative generation increases 9.955 → 11.120 TWh (+11.7%).
- **Annual CO_2 barely changes**
  The RGGI cap binds from 2027 onward in both scenarios, holding annual emissions near 6.17 Mt.
- **Timing versus annual total**
  CPS changes when fossil generation occurs; RGGI constrains the annual total.

Full certificate compliance therefore does not imply the same fossil displacement during clean-peak hours.

Source: Dai & Yao (2026). The modeled RGGI cap binds from 2027 onward in both scenarios.

Recalibration bridges the gap, but cannot replace physical deployment

1 Feasibility improves

Full CPS compliance is maintained while modeled system cost falls by 3.7%.

2 Physical performance weakens

The system builds 729 MW less new storage, delivers 1.82 TWh less eligible discharge, and uses 1.17 TWh more fossil generation during CPS hours.

3 Track certificates and physical outcomes together

Project multipliers can bridge near-term delivery constraints, but policymakers should also monitor installed capacity, physical discharge, and fossil displacement.

Scope note: the model does not explicitly simulate permitting, interconnection, procurement, or construction delays, and it assumes all candidate batteries qualify for the 2× multiplier.

Interpretation based on Dai & Yao (2026). Results reflect the combined change in standards and NTRM treatment.

Thank you

Questions and discussion

Cheaper certificate compliance does not guarantee the same physical clean-peak outcome.

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